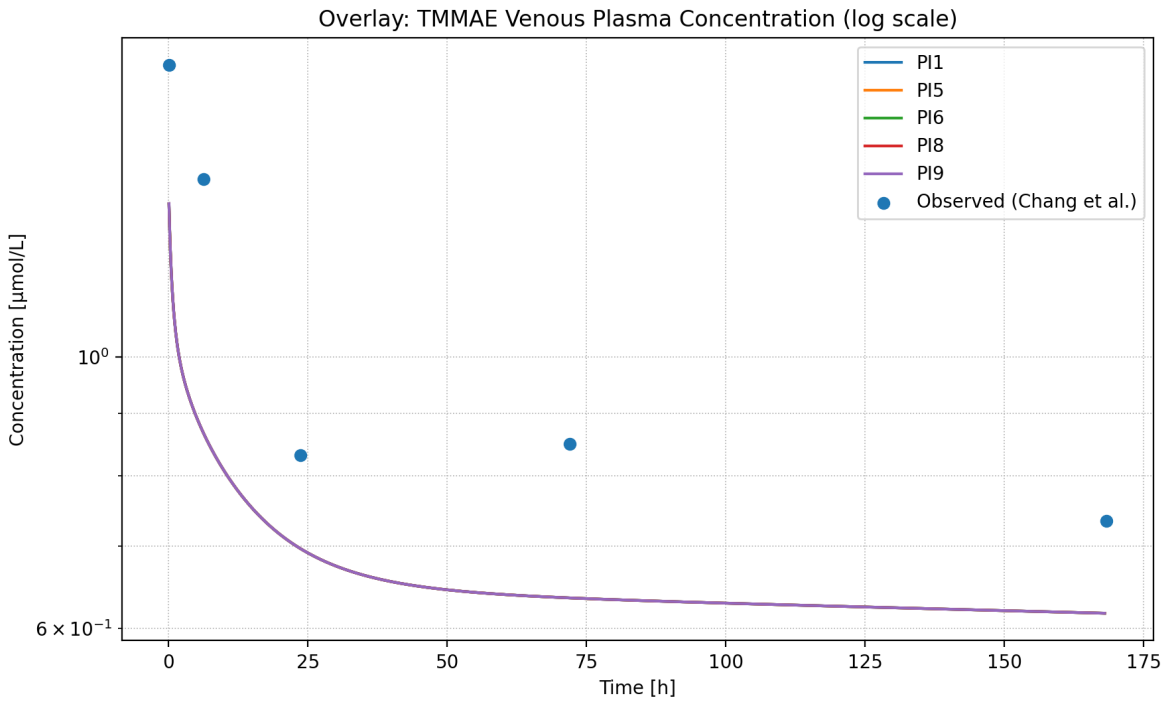
Global Optimization Summary (Random Starts)

Objective: Total Error (all runs shown reached the same best value: **0.73**)

Optimization results across random starts

Run #	K_d [µmol L^{-1}]	K_d**(FcRn), endosomal** [µmol L^{-1}]	**Reference conc.** [µmol L^{-1}]
1	9756.770	$3.150 \cdot 10^{-7}$	$1.430 \cdot 10^{-3}$
2	9728.660	$3.460 \cdot 10^{-7}$	$3.920 \cdot 10^{-3}$
3	9483.820	$2.090 \cdot 10^{-7}$	$4.890 \cdot 10^{-3}$
4	9321.530	$8.060 \cdot 10^{-7}$	$1.000 \cdot 10^{-2}$
5	7214.690	$5.740 \cdot 10^{-7}$	$9.440 \cdot 10^{-5}$
6	9795.130	$5.050 \cdot 10^{-7}$	$2.390 \cdot 10^{-3}$
7	9666.190	$4.010 \cdot 10^{-7}$	$1.030 \cdot 10^{-4}$
8	9242.190	$1.100 \cdot 10^{-7}$	$3.580 \cdot 10^{-3}$
9	7805.050	$4.090 \cdot 10^{-7}$	$9.980 \cdot 10^{-5}$
10	8352.370	$4.640 \cdot 10^{-7}$	$4.810 \cdot 10^{-4}$



Overlay of predicted concentration–time profiles (log-scale) for different parameter-identification runs. X-axis: time [h]. Y-axis: concentration [µmol L^{-1}] on a logarithmic scale. Curves correspond to PI1, PI5, PI6, PI8, PI9; points show observed data (Chang et al.).

Interpretation of the overlay figure

- The predicted curves from different parameter-identification runs are **visually indistinguishable** (they lie on top of each other).
- This indicates **practical non-identifiability**: multiple parameter combinations produce the same model predictions for the fitted observable.
- Consequently, parameters can drift (often toward bounds) **without affecting the fit**, so bound-hitting behavior (e.g., K_d at max) should be interpreted as **lack of constraint by the data/model output**, not necessarily as a unique biological truth.

Key takeaways (what these results imply)

- **Same best error across starts (≈ 0.73):** suggests a broad optimum / many parameter sets yield similar fit quality.
- **K_d consistently near the upper bound (max = 10,000 $\mu\text{mol}\,\text{L}^{-1}$):** optimizer prefers *very weak binding* (large K_d) *within the allowed bounds.*
- **Endosomal FcRn K_d collapses to very small values ($\sim 10^{-7}$ $\mu\text{mol}\,\text{L}^{-1}$):** optimizer prefers *very strong* endosomal FcRn binding (or the parameter is weakly informed and can go extreme without penalty).
- **Reference concentration varies widely with no change in error:** indicates trade-offs / non-identifiability (multiple combinations fit equally well).

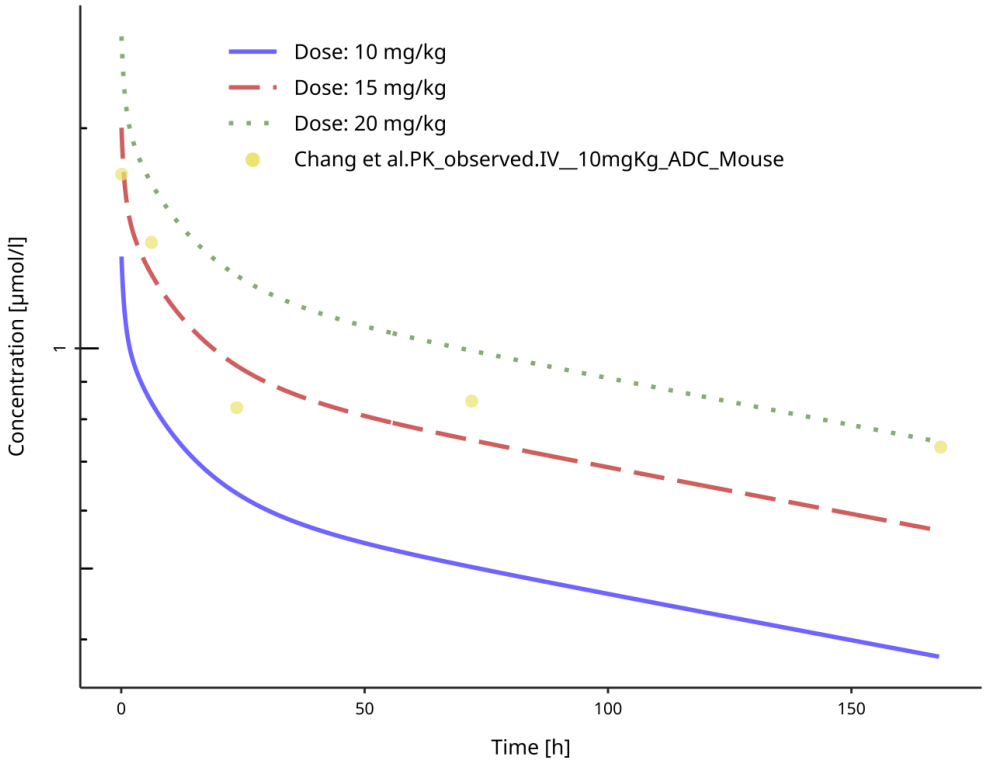


Figure 1: Model response to different dose values.

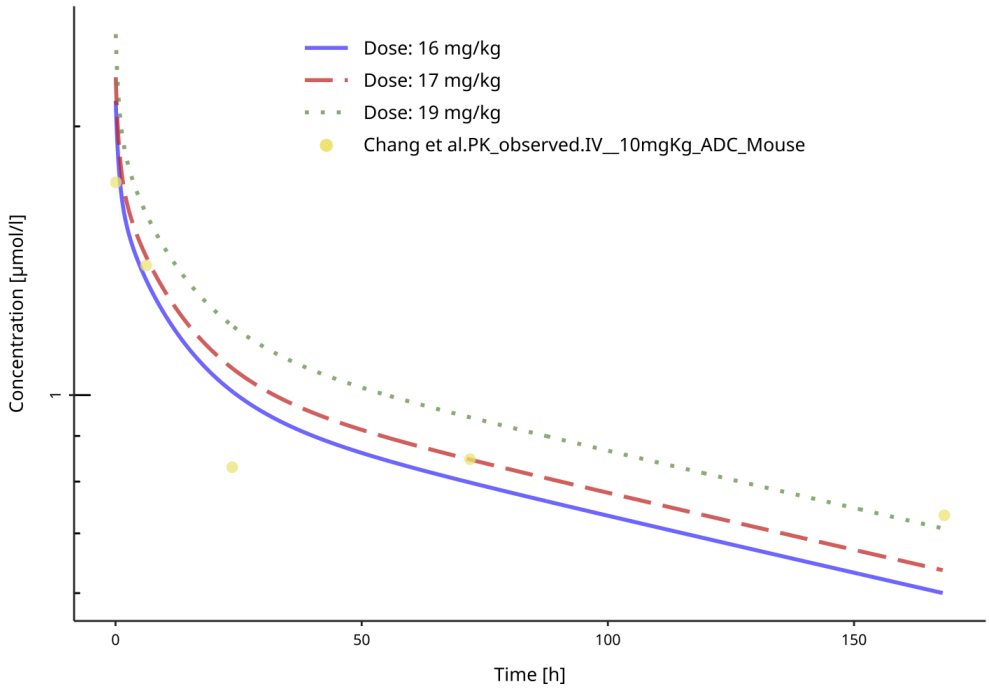


Figure 2: Model response to different dose values.

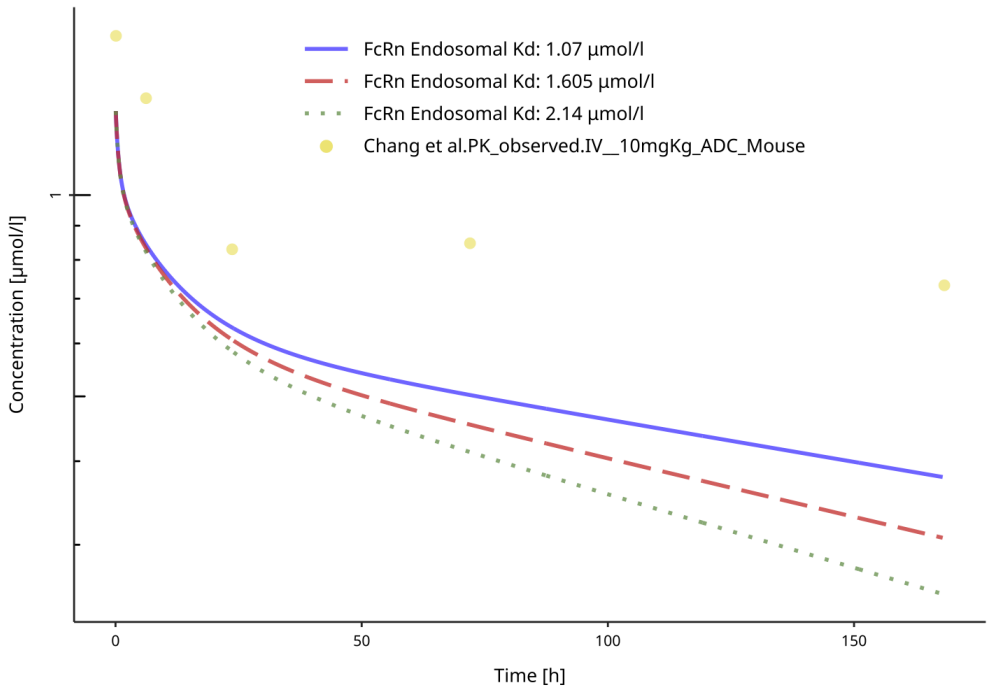


Figure 3: Model response to different FcRn endosomal K_d values.

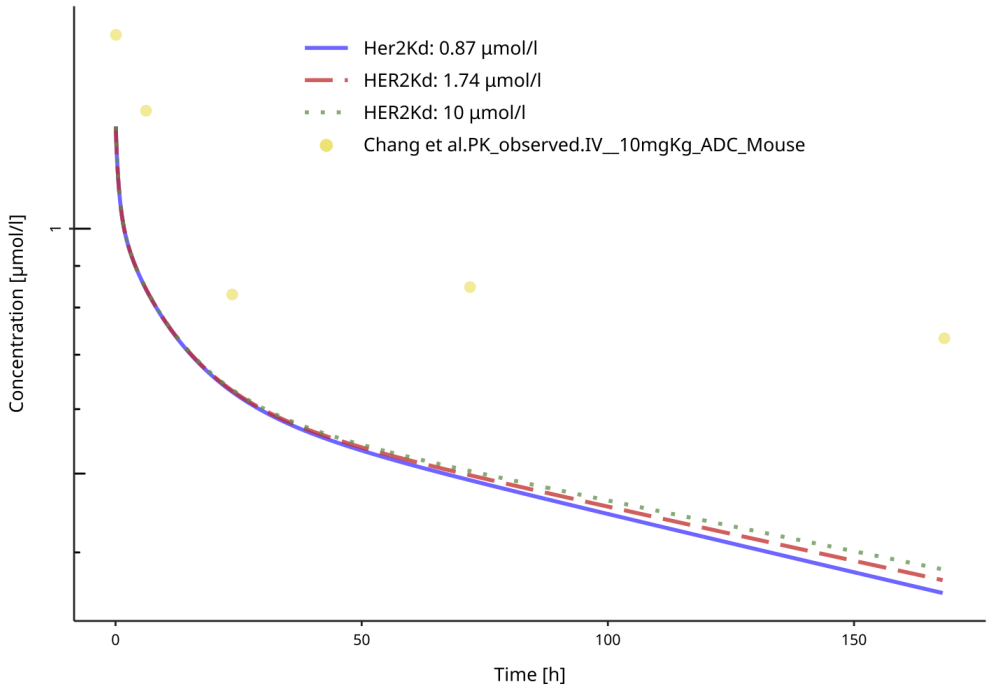


Figure 4: Model response to different HER2 binding K_d values.

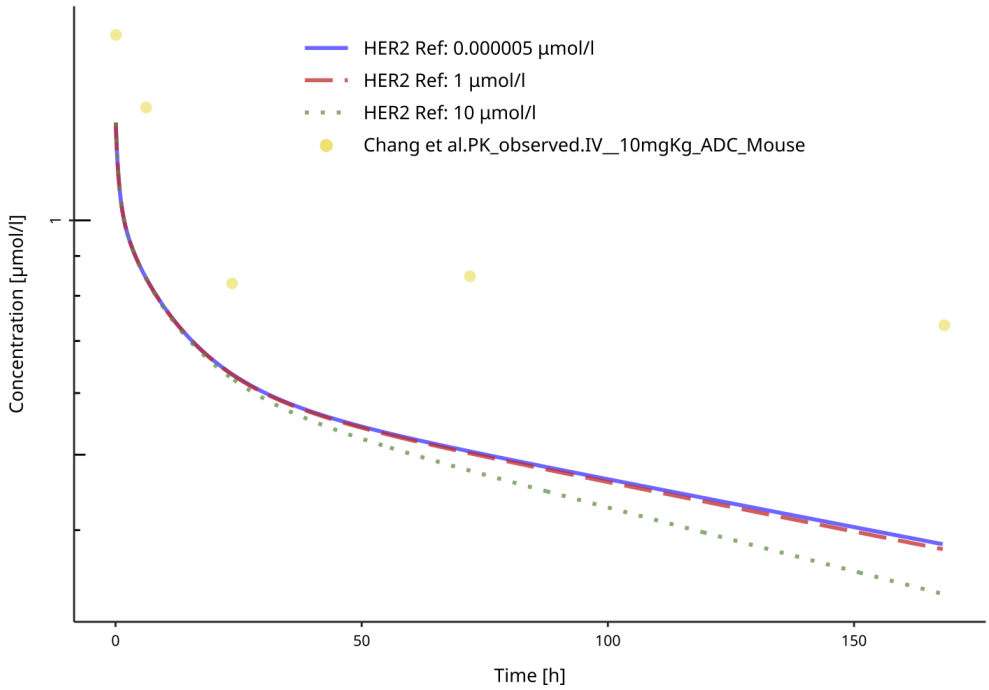


Figure 5: Model response to different HER2 reference concentration values.

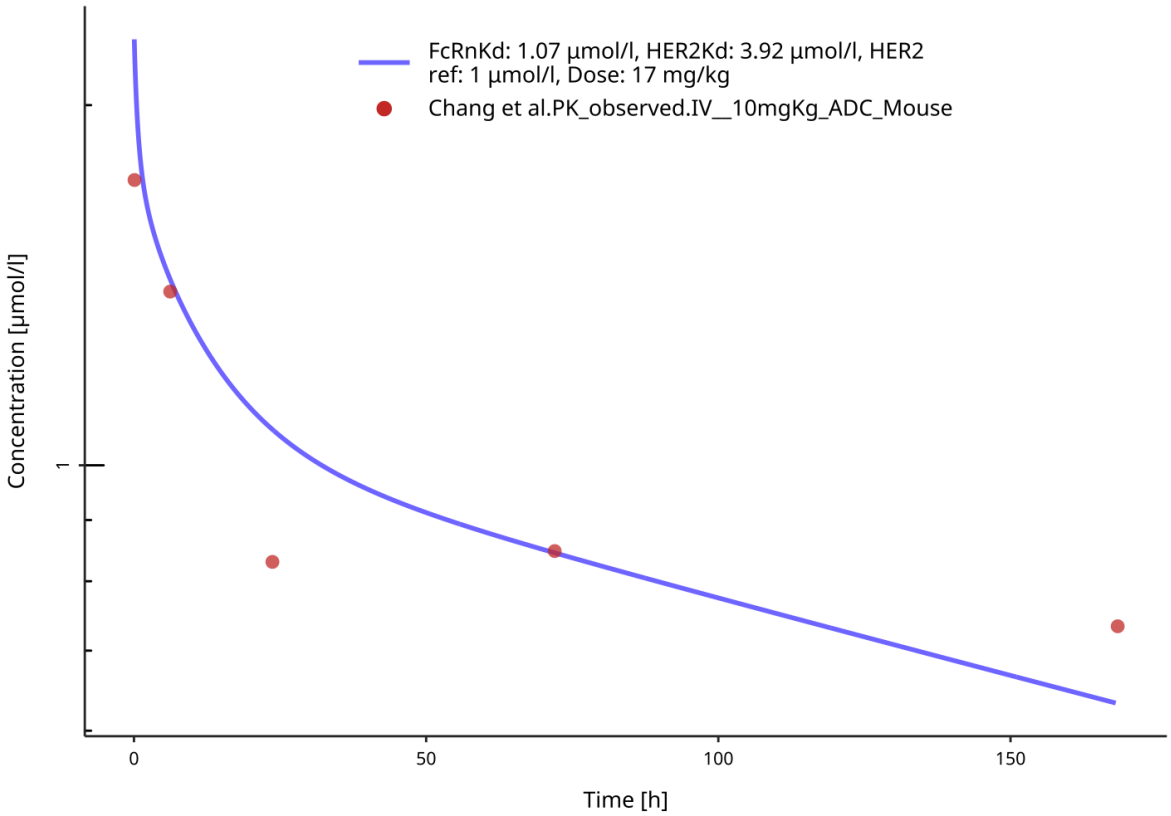


Figure 6: Model response to the optimized parameter values.

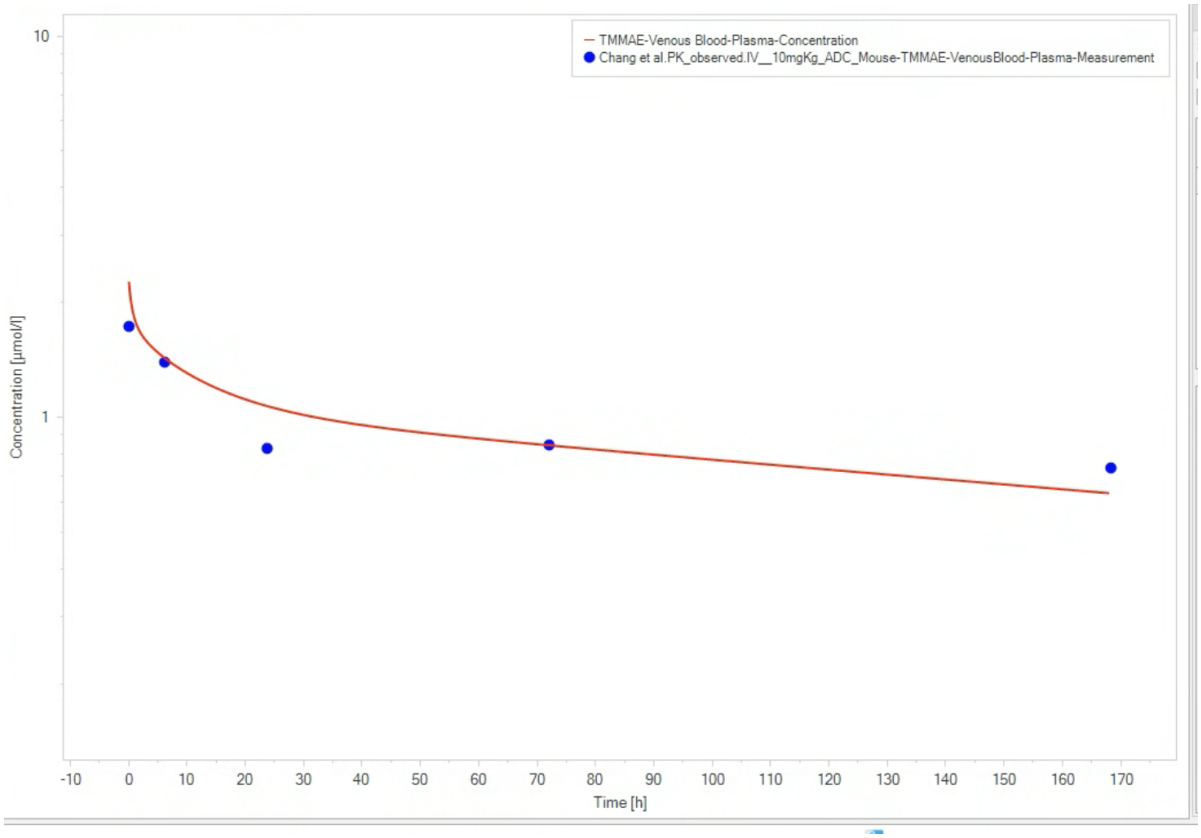


Figure 7: Model response to the optimized parameter values on PKSim